

PADANY, Russian Federation

WMO: 226190

Lat: 63.2668N

Lon: 33.4101E

Elev: 427

StdP: 14.47

Time Zone: 3.00 (E03)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.3	-12.9	-22.8	1.6	-18.4	-17.1	2.2	-13.0	22.5	20.8	18.8	18.8	3.4	300	0.632

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	78.3	65.5	75.2	64.0	72.0	62.5	68.1	75.6	65.8	72.5	63.6	70.5	5.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.2	94.6	72.6	63.0	87.5	70.7	60.4	79.7	68.0	32.6	75.6	30.9	72.1	29.1	71.2	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min		Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.9	17.8	15.2		-23.4	83.4	6.3	4.3	-27.9	86.5	-31.6	89.0	-35.2	91.5	-39.8	94.6	
				-23.6	69.6	6.3	3.0	-28.1	71.7	-31.8	73.5	-35.3	75.1	-39.9	77.3	
				DB												
				WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.6	12.3	15.7	23.2	34.5	47.2	55.6	62.2	59.4	50.8	37.9	29.8	20.9	
	DBStd	19.07	13.68	13.53	10.05	7.52	8.70	7.16	6.95	6.04	5.66	6.37	7.22	12.39	
	HDD50	5560	1170	960	832	467	159	24	3	3	59	375	606	902	
	HDD65	10090	1635	1380	1297	916	554	293	136	191	426	839	1056	1367	
	CDD50	1027	0	0	0	0	72	193	382	296	82	2	0	0	
	CDD65	84	0	0	0	0	3	11	51	19	0	0	0	0	
	CDH74	400	0	0	0	0	32	59	232	77	0	0	0	0	
	CDH80	59	0	0	0	0	7	7	31	14	0	0	0	0	
Wind	WSAvg	7.0	6.8	6.8	7.5	6.7	6.8	6.6	5.8	6.4	7.0	8.4	8.1	7.5	
Precipitation	PrecAvg	21.0	1.0	0.8	0.9	1.2	1.5	2.4	2.6	3.2	2.4	2.1	1.5	1.2	
	PrecMax	32.6	2.0	2.2	2.4	3.0	3.5	5.7	7.2	8.3	5.8	4.1	3.0	2.4	
	PrecMin	14.5	0.2	0.1	0.1	0.2	0.2	0.6	0.5	0.5	0.4	0.0	0.2	0.3	
	PrecStd	3.8	0.4	0.4	0.5	0.7	0.7	1.2	1.3	1.7	1.1	0.9	0.7	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.5	38.8	47.0	60.0	78.2	80.2	83.6	81.5	67.6	54.9	47.4	40.0	
		MCWB	33.3	36.1	37.7	46.9	59.3	63.3	68.5	66.7	59.3	50.4	45.2	38.4	
	2%	DB	33.7	36.1	40.7	51.5	70.0	74.8	79.5	75.7	63.5	51.1	43.1	37.1	
		MCWB	32.6	33.9	34.5	41.1	54.5	60.8	66.8	64.5	56.8	47.8	41.2	36.0	
	5%	DB	32.3	34.3	37.7	47.8	65.3	71.1	76.4	72.2	60.8	48.8	40.7	35.1	
		MCWB	31.6	32.6	33.5	39.6	52.5	61.4	65.7	62.4	55.3	46.0	39.3	33.9	
	10%	DB	30.9	32.8	35.5	44.5	61.0	67.2	73.6	69.3	58.6	46.6	38.1	33.6	
		MCWB	30.1	31.7	32.7	38.2	50.7	58.8	64.8	61.2	54.1	44.3	36.8	32.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.0	36.3	38.5	47.0	61.2	68.0	71.1	69.8	60.4	51.5	45.2	39.0
			MCDB	34.7	38.2	44.2	59.3	73.2	73.9	80.0	77.5	65.3	54.5	47.3	40.1
2%		WB	32.8	34.2	35.7	42.9	56.8	64.4	69.0	66.1	58.2	48.8	41.7	36.1	
		MCDB	33.4	35.7	39.3	49.4	66.3	72.9	76.6	73.1	61.9	50.3	43.0	36.9	
5%		WB	31.7	32.9	34.2	40.6	53.9	61.7	67.4	63.7	56.4	46.2	39.4	34.1	
		MCDB	32.4	34.2	36.9	46.0	63.2	69.6	74.7	69.8	59.7	47.9	40.7	34.8	
10%		WB	30.1	31.7	32.9	38.9	51.0	58.8	65.3	61.8	54.8	44.3	37.1	32.7	
		MCDB	31.1	33.0	35.4	43.8	59.6	65.7	72.1	68.3	57.9	46.6	38.0	33.5	
Mean Daily Temperature Range	5% DB	MDBR	9.8	10.7	13.0	13.1	14.0	12.5	11.7	11.3	9.0	6.6	5.9	7.7	
		MCDBR	6.2	8.4	11.6	17.3	21.4	17.8	16.4	15.6	12.6	9.4	8.4	6.4	
		MCWBR	6.0	7.6	8.1	10.0	10.1	8.7	7.9	7.7	7.5	7.0	8.0	6.2	
	5% WB	MCDBR	5.7	8.2	9.9	14.1	19.1	15.4	14.3	13.8	10.5	8.2	8.2	6.3	
		MCWBR	5.5	7.6	7.8	9.3	11.2	8.8	7.8	8.2	7.2	7.4	7.8	6.2	
Clear-Sky Solar Irradiance	taub	0.226	0.281	0.309	0.345	0.350	0.343	0.374	0.355	0.329	0.306	0.236	0.205		
	taud	1.987	2.097	2.158	2.288	2.447	2.496	2.425	2.455	2.513	2.394	2.069	1.956		
	Ebn at Noon	120	201	246	265	275	279	265	260	246	200	116	61		
	Edh at Noon	10	21	30	33	31	30	31	28	22	16	9	5		
All-Sky Solar Radiation	RadAvg	60	246	712	1181	1500	1664	1558	1164	658	246	62	25		
	RadStd	7	49	80	106	162	122	169	118	70	33	11	3		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)	+1.17	N/A	N/A	N/A	N/A	N/A	N/A	-409	N/A	N/A

Nomenclature: See separate page